

MBB data story — plot highlights (talk track)

Deck:  · read **top-left** → **bottom-right**
Reigning HERO lock: 09–21 · last-ps · EW16 poolq_LOO · min20 · mg10 · ever-Y
Use one sentence per panel when screening a domain; two if Alex asks “why is this here?”

Row 1 — Who is in the pond?

1 · Cohort

Highlight: This is the **decision universe** — one row per athlete at final college season, POST-QC filters, and the outcome we care about (ever drafted).
Say: “N = 22,795 last-ps rows, 615 ever-drafted — everything below is about whether **peer context** predicts draft beyond **own ability**.”

2 · $\hat{A}_i$ and T_j

Highlight: Separates **individual talent** ($\hat{A}$) from **team-level realized talent** (T_j , includes self).
Say: “Draftees sit right-shifted on ability; team talent has its own distribution. HERO’s environment axis is **LOO peers**, not raw T_j — this panel sets up *who* vs *where*.”

3 · poolq_LOO distribution (3a histogram · 3b ECDF)

Highlight: Shows the **support** of the peer-context variable — where LOO teammate quality lives on the z-scale, and whether drafted players come from systematically different ponds.
3a (histogram, no winsor): Raw LOO support — full spread without 1–99% clip artifacts at the tails. Blue = all last-ps rows; purple +DFT overlay = rows on teams with any ever-drafted athlete.
3b (ECDF): Same comparison in cumulative form. If the **+DFT curve sits above and left** of the full-panel curve, draftees systematically come from **higher peer ponds** — the entire drafted subpopulation is shifted toward better LOO, not just a histogram bump in one bin.
Value: Before any binning story, you need LOO to be **non-degenerate** and to see that draftees **sort into better teammate contexts** on average — a descriptive prerequisite for HERO, not the hero curve itself.
Say: “Left: where LOO mass lives. Right: drafted players’ ponds are **stochastically better** — the +DFT curve rises faster. That’s the input side before we bin draft *rates*.”
Not: An outcome curve — descriptive **input** distribution only (reigning HERO lock still winsorizes LOO at fit time; this panel is porch truth-telling).

Row 2 — Geometry and who carries draft mass

4 · Draft mass vs $\hat{A}$ (ECDF)

Highlight: Where **draft picks come from in ability space** — cumulative share of all drafts accounted for by ability tiers.
Value: Justifies **fixing $\hat{A}$ bands** later (CCT, elite pond): most draft mass is **not** uniform across ability — a thin slice of $\hat{A}$ carries disproportionate draft probability. Without this, “fix $\hat{A}$ at $z \in [2,3]$ ” or “top 7%” looks arbitrary.
Say: “Most draft capital concentrates in the upper ability tail — so when we **hold ability fixed** in panels 7–8, we’re zooming where the **marginal draft decision** actually happens, not the whole roster.”
Pairs with: Panel 2 (marginal $\hat{A}$) but Panel 4 answers **draft-weighted** importance, not headcount.

5 · Team interval overlap

Highlight: **Sorting overlap** — how much team talent windows stack on the same roster (assortment / congestion geometry).
Say: “Teams aren’t isolated talent bins — intervals overlap. That’s why LOO (who’s left when you remove self) matters more than ‘my team’s average’ alone.”

6 · Team roster size $|T_j|$

Highlight: **Pond size** — how many meaningful minutes-players define each team-season LOO pool.
Say: “LOO is computed over ~12–13 rotation players per team-season (min20). Peer context is a **real roster**, not an abstract league average.”

Row 3 — Conditional stories (all LOO-axis)

7 · CCT P1 — fixed $\hat{A} \ z \in [2,3]$ · QTL16 · LOO

Highlight: **Squid vs Jackal within a matched ability band** — mid-pond vs top-pond LOO ventiles, holding **PPM $z \in [2, 3]$** fixed (upper tail with a ceiling at $z = 3$, not a ventile label).

Knob	Panel 7
Who	Fixed z window — PPM $z \in [2.0, 3.0]$ (within-season z; not “ventiles 2–3”)
Share	~2.4% of +DFT all-ps panel (11–21 deck: n = 425); c top 7% (everyone in [2, 3] is above the top-7% cut $z \geq 1.5$)
Binning	QTL16 on poolq_LOO <i>within that band</i>
Question	Among similarly talented players, do mid-LOO ponds draft more than top-LOO ponds? (CCT signature)
Read	Squid (mid) vs Jackal (top) draft rates; CCT = YES if Squid > Jackal
Optional	Within-band LPM on raw poolq_LOO: $\beta_2 < 0$ (concave) — supports CCT but not the prespecified test (deck: $\beta_2 \approx -0.49$)

Say: “Panel 7 is **Act II conditional congestion** on a **narrow z slice** [2, 3] — roughly the top ~2–3% of ability, capped below the extreme $z > 3$ tail. Same LOO axis, but a **tighter $\hat{A}$ gate** than panel 8’s top 7%. Classic CCT Squid–Jackal test — not the full HERO curve.”

Not: Top-7% elite gate; not piecewise tail bins; not the reigning 09–21 last-ps aperture. Deck uses **11–21 +DFT** (sharper Squid–Jackal gap). **09–21 still CCT=YES** (35% vs 28%) — side-by-side: [_compare_0921/COMPARE_panel7_CCT_z23_9_21_vs_11_21.png](#).

8 · Elite pond LOO — top 7% $\hat{A}$ · PW 4+7 · LOO

Highlight: Elite players only — draft rate vs **peer context** with **piecewise tail** binning on LOO (coarse low + fine high LOO).

Knob	Panel 8
Who	Top 7% $\hat{A}$ (pooled percentile cut on PPM z; deck 11–21: $z \geq 1.49$)
Share	7.0% of +DFT all-ps panel (11–21 deck: n = 1,237) — wider than panel 7
Binning	PW 4+7 on poolq_LOO (piecewise tail — not QTL, not EW16)
Question	Among elites, does draft rate rise then fall in the highest-LOO tail?
Read	Plateau → peak → downturn in top LOO bins (+DFT 11–21: 25% → 14% tail)

Say: “Panel 8 is the **elite-pond keeper** — same LOO axis as HERO, but the explicit **top 7% $\hat{A}$ gate** (broader than panel 7’s $z \in [2, 3]$ slice) and PW 4+7 bins tuned for the **elite tail dip**.”

Compare windows: [_compare_0921/COMPARE_panel8_elite_top7_9_21_vs_11_21.png](#) — downturn visible in both (09–21: 26% → 17% tail; 11–21 deck: 25% → 14%).

7 vs 8 — one-liner contrast

Panel 7: Among **very-high- $\hat{A}$** players ($z \in [2, 3]$, ~top 2–3% · capped at $z = 3$), do **middle** peer ponds beat **top** peer ponds? → **CCT / Squid vs Jackal**.

Panel 8: Among **elite** players (**top 7% $\hat{A}$** , $z \geq \sim 1.5$), what happens at the **extreme high-LOO** tail? → **Elite pond downturn**.

Same **LOO x-axis family**; different **$\hat{A}$ gate** (panel 7 is a **narrower, higher** subset of panel 8), **binning**, and **question**.

$\hat{A}$ gate	Definition	+DFT 11–21 (deck)	Reigning 09–21 last-ps ALLT
Panel 7	$z \in [2, 3]$	425 rows (2.4%)	733 rows (3.2%)
Panel 8	Top 7%	1,237 rows (7.0%) · $z \geq 1.49$	1,596 rows (7.0%) · $z \geq 1.65$

9 · HERO (Pass A · EW16 · LOO) — finale

Highlight: Full-panel environment curve — draft rate vs poolq_LOO for everyone passing filters (not ability-fixed).

Say: “Panel 9 is the **reigning HERO** — middle rise, flat elite LOO tail on POST-QC panel; $\beta_2 \approx +0.0017$ (not concave on this spec). Panels 7–8 explain *where* in $\hat{A}$ -space the interesting conditional shapes live.”

30-second domain screen (Alex)

- Population sane? → 1, 2, 6
- Environment variable sane? → 3
- Draft mass justifies zoom bands? → 4
- Assortment / overlap? → 5
- Act II congestion? → 7
- Elite tail on LOO? → 8
- HERO shape? → 9

Verdict strip: HERO middle rise + flat/dip elite tail; CCT band shows Squid vs Jackal; elite pond shows LOO tail drop among top 7%.

Footnotes for honest slides

- Panels **7–8**: 2011–21 +DFT (deck); panel **9**: 09–21 last-ps ALLT (reigning lock). Row 3 is **LOO-aligned**; window/pop differ slightly.
- 09 vs 11–21 compares** (same spec, +DFT all-ps): [_compare_0921/COMPARE_panel7_*.png](#) , [COMPARE_panel8_*.png](#) ; source PNGs [*_9_21.png](#) / [*_11_21.png](#) .
- Panel 7 **$\hat{A}$ gate** = fixed z window [2, 3], not ventiles; **c top 7%**. Panel 8 = **percentile top 7%**.
- Panel 8 binning = **PW 4+7**, not QTL16 or EW16. Panel 7 binning = **QTL16** within the z band.

- Panel 7 β_2 (within-band LPM on poolq_LOO) is optional annotation only; primary read = **Squid vs Jackal / CCT=YES**. Panel 9 β_2 is full-panel HERO curvature — different sample and question.
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